EEEE4064
ADVANCED CONTROL SYSTEM DESIGN

Lecture 7
Controller Design Using Pole Placement -1





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Lecture Outline

01 Pole Placement Theory

02 Pole Placement Design

03 Transformation Matrix

04 Direct Substitution

05 Ackermann's Formula

Pole Placement Theory

CONCEPT

- The **pole placement** method, also known as the **full state feedback** method, is somewhat similar to the root-locus method in that we place closed-loop poles at desired locations.
- The basic difference is that in the root-locus design we place only the **dominant** closed-loop poles at the desired locations, while in the pole placement design we place **all** closed-loop poles at desired locations.

POLES LOCATIONS

The first step in the pole-placement design approach is to choose the locations of the desired closed-loop poles. The most frequently used approach is to choose such poles based on experience in the root-locus design. This is done by placing a **dominant pair** of closed-loop poles and choosing **other poles** so that they are **far to the left** of the dominant closed-loop poles.

Pole Placement Theory

POLES LOCATIONS

If we place the dominant closed-loop poles **far** from the $j\omega$ axis, so that the system response becomes **very fast**, the signals in the system become **very large**, with the result that the system may become **nonlinear**. This should be avoided.

ARBITRARY PLACEMENT

Arbitrary pole placement for a given system is possible if and only if the system is completely state controllable. The best practices for such an approach include considering the following:

01 Based on experience in the root-locus design, place a dominant pair of closed-loop poles. Then, place other poles far to the left of the dominant closed-loop poles.

Pole Placement Theory

02 Too far closed-loop poles to the left cause faster response, and higher gains and wider frequency band. This will also increase the effect of measurement noises.

03 Too far dominant closed-loop poles to the left drive the system response into nonlinear regions.

04 Too far dominant closed-loop poles to the left require higher amount of control energy. Thus, there is a balance between acceptable system response and the amount of control energy required.

Pole Placement Design

REGULATORS & CONTROLLERS

Systems that include controllers can be divided into two categories: **regulator systems** (where the reference input is constant, including zero) and **control systems** (where the reference input is time varying). In this lecture, we will consider regulator systems.

REGULATORS DESIGN

The process of choosing the appropriate state feedback gain matrix ($\mathbf{K}$) such that the system poles are re-located to the desired locations is known as the pole placement problem for state space equations.

We consider the following:

- All the state variables are measurable and are available for feedback.
- The system is completely state controllable.
- The input is constant, including zero, (regulator system).

Pole Placement Design

RELATIONSHIPS

Consider the completely state controllable system:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$
$$y = \mathbf{C}\mathbf{x} + Du$$

Consider the control signal u:

$$u = -\mathbf{K}\mathbf{x}$$

$\mathbf{K}$ is called the state feedback gain matrix. By substituting u:

$$\dot{\mathbf{x}}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t)$$

The solution of this equation is given by:

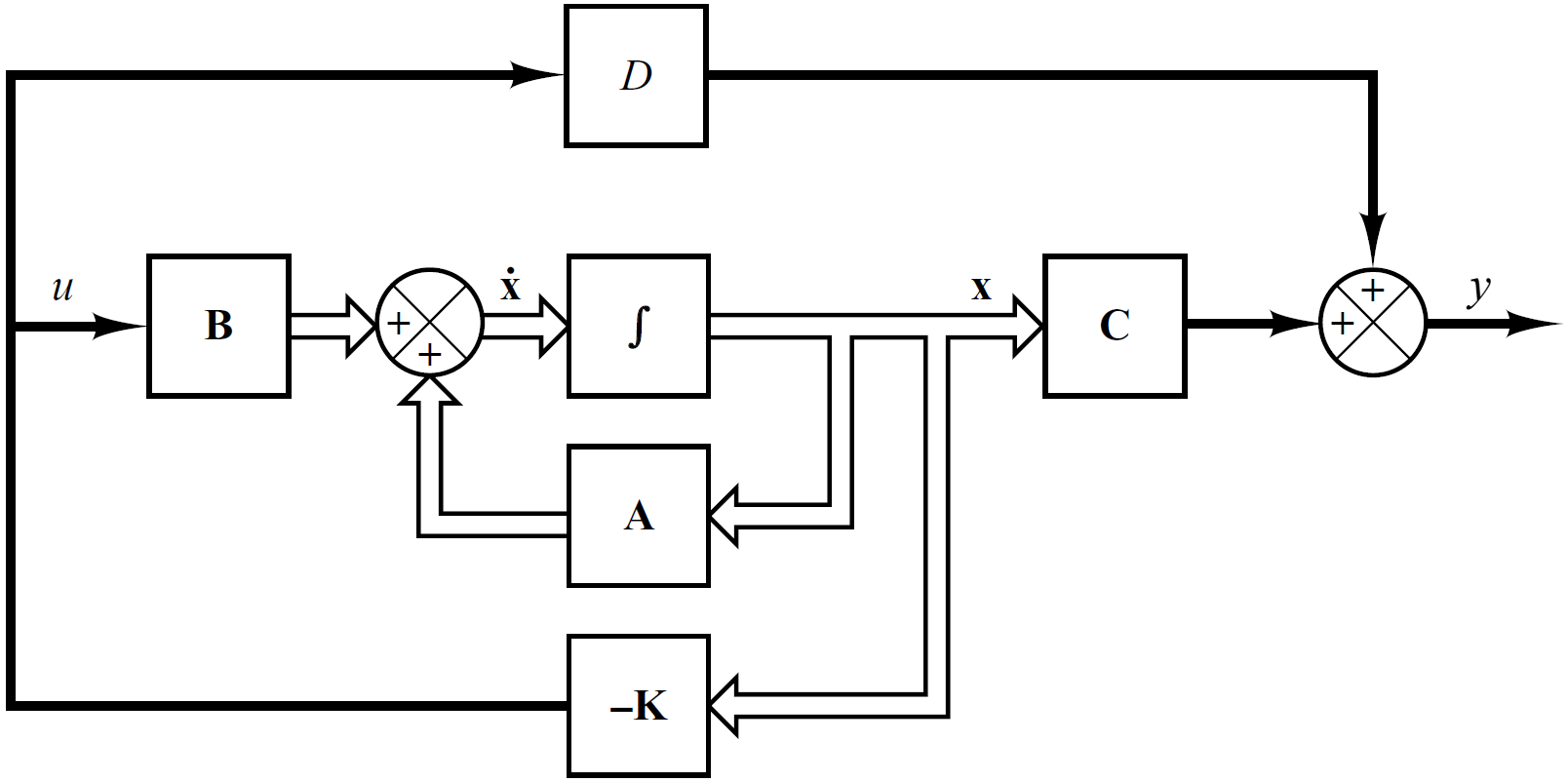
$$\mathbf{x}(t) = e^{(\mathbf{A}-\mathbf{B}\mathbf{K})t}\mathbf{x}(0)$$

$\mathbf{x}(0)$ is the initial state caused by external disturbances.

Pole Placement Design

REGULATOR'S STRUCTURE

The previous relationships can be represented as follows:



Notes on Pole Placement Design

01 The stability and transient response characteristics are determined by the eigenvalues of matrix $\mathbf{A-BK}$.

02 If the matrix $\mathbf{K}$ is chosen properly, the system can be made to be asymptotically stable. It can drive the system state from any initial value to 0.

03 If the regulator poles are placed in the left-half of the s-plane, then $\mathbf{x}(t)$ approaches 0 as t approaches infinity.

04 Pole placement can be done using several methods, such as the **transformation matrix**, **direct substitution**, and **Ackermann's formula**.

Transformation Matrix

INTRODUCTION

The transformation matrix method is also known as the controllable canonical form method. It involves the following steps:

01 Check the state controllability condition of the system.

02 Determine the values of $a_1, a_2, \ldots, a_n$ from the characteristic equation of the open-loop system:

$$|s\mathbf{I}-\mathbf{A}|=s^n+a_1s^{n-1}+\cdots+a_{n-1}s+a_n$$

03 Using the desired eigenvalues (desired closed-loop poles), $\mu_1, \mu_2, \ldots \mu_n$, write the desired characteristic polynomial and determine the values of $\alpha_1, \alpha_2, \ldots, \alpha_n$.

$$(s-\mu_1)(s-\mu_2)\cdots(s-\mu_n)=s^n+\alpha_1s^{n-1}+\cdots+\alpha_{n-1}s+\alpha_n=0$$

Transformation Matrix

04

Determine the transformation matrix $\mathbf{T}$ that transforms the system state equation into the **controllable canonical form**. If the given system equation is already in the controllable canonical form, then $\mathbf{T} = \mathbf{I}$.

$$\mathbf{T} = \mathbf{MW} = \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix} \begin{bmatrix} a_{n-1} & a_{n-2} & \cdots & a_1 & 1 \\ a_{n-2} & a_{n-3} & \cdots & 1 & 0 \\ \cdot & \cdot & & \cdot & \cdot \\ a_1 & 1 & \cdots & 0 & 0 \\ 1 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

$$\mathbf{TAT}^{-1} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \qquad \mathbf{T}^{-1}\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \\ 1 \end{bmatrix}$$

Transformation Matrix

05

The required state feedback gain matrix $\mathbf{K}$ can be determined from the following equation:

$$\mathbf{K} = \begin{bmatrix} \alpha_n - a_n & \alpha_{n-1} - a_{n-1} & \dots & \alpha_1 - a_1 \end{bmatrix} \mathbf{T}^{-1}$$

Direct Substitution

INTRODUCTION

Direct substitution of matrix $\mathbf{K}$ into the desired characteristic polynomial may be simpler If the system is of low order ($n \leq 3$). otherwise, it become very tedious. It involves the following steps:

01 Check the state controllability condition of the system.

02 For example, if $n = 3$, then write the state feedback gain matrix $\mathbf{K}$ as:

$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}$$

03 Substitute this $\mathbf{K}$ matrix into the desired characteristic polynomial to determine the values of k_1, k_2, and, k_3:

$$|s\mathbf{I} - \mathbf{A} + \mathbf{BK}| = (s - \mu_1)(s - \mu_2)(s - \mu_3)$$

Ackermann's Formula

INTRODUCTION

This method was proposed by Jurgen Ackermann. It involves the following steps:

01 Check the state controllability condition of the system.

02 Find the matrix $\phi(\mathbf{A})$:

$$\phi(\mathbf{A}) = \mathbf{A}^n + \alpha_1 \mathbf{A}^{n-1} + \cdots + \alpha_{n-1}\mathbf{A} + \alpha_n \mathbf{I}$$

03 Design $\mathbf{K}$ using the following formula:

$$\mathbf{K} = \begin{bmatrix} 0 & 0 & \cdots & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{B} & \mathbf{AB} & \cdots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}^{-1} \phi(\mathbf{A})$$

Regulator Design

EXAMPLE

Determine the state feedback gain matrix $\mathbf{K}$ of the regulator system shown in Slide 8 if the plant is given by:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Assume that the desired closed-loop poles are at:

$$s = -2 + j4, \quad s = -2 - j4, \quad s = -10$$

Use different methods to find the answer. Then, investigate the response of the system using MATLAB when the initial conditions are $\mathbf{x}(0) = [1 \;\; 0 \;\; 0]$.

Regulator Design

Test the state controllability, which will be used for all methods:

$$|\mathbf{M}| = \left|\mathbf{B} \quad \mathbf{AB} \quad \mathbf{A}^2\mathbf{B}\right| = \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 31 \end{vmatrix} = -1$$

Thus, the system is completely state controllable.

Transformation Matrix Method:

$$|\,s\mathbf{I} - \mathbf{A}\,| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 5 & s+6 \end{vmatrix} = s^3 + 6s^2 + 5s + 1 = s^3 + a_1 s^2 + a_2 s + a_3 = 0$$

$$a_1 = 6, \quad a_2 = 5, \quad a_3 = 1$$

$$(s+2-j4)(s+2+j4)(s+10) = s^3 + 14s^2 + 60s + 200 = s^3 + \alpha_1 s^2 + \alpha_2 s + \alpha_3 = 0$$

$$\alpha_1 = 14, \quad \alpha_2 = 60, \quad \alpha_3 = 200$$

Regulator Design

The system is in the controllable canonical form. Thus, $\mathbf{T} = \mathbf{T}^{-1} = \mathbf{I}$

$$\mathbf{K} = \begin{bmatrix} \alpha_3 - a_3 & \alpha_2 - a_2 & \alpha_1 - a_1 \end{bmatrix} \mathbf{T}^{-1} = \begin{bmatrix} 200-1 & 60-5 & 14-6 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 199 & 55 & 8 \end{bmatrix}$$

Direct Substitution Method:

$$\mathbf{K} = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix}$$

$$|\, s\mathbf{I} - \mathbf{A} + \mathbf{BK} \,| = \left| \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} \right| = \begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1+k_1 & 5+k_2 & s+6+k_3 \end{vmatrix}$$

$$|\, s\mathbf{I} - \mathbf{A} + \mathbf{BK} \,| = s^3 + (6+k_3)s^2 + (5+k_2)s + 1 + k_1 = 0$$

$$s^3 + (6+k_3)s^2 + (5+k_2)s + 1 + k_1 = s^3 + 14s^2 + 60s + 200$$

Regulator Design

$$6+k_3=14,\quad 5+k_2=60,\quad 1+k_1=200 \Rightarrow \mathbf{K}=\begin{bmatrix}199 & 55 & 8\end{bmatrix}$$

Ackermann's Formula:

$$\phi(\mathbf{A})=\mathbf{A}^3+\alpha_1\mathbf{A}^2+\alpha_2\mathbf{A}+\alpha_3\mathbf{I}=\mathbf{A}^3+14\mathbf{A}^2+60\mathbf{A}+200\mathbf{I}$$

$$\phi(\mathrm{A})=\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -5 & -6\end{bmatrix}^3+14\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -5 & -6\end{bmatrix}^2+60\begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ -1 & -5 & -6\end{bmatrix}+200\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}199 & 55 & 8\\ -8 & 159 & 7\\ -7 & -43 & 117\end{bmatrix}$$

$$\begin{bmatrix}\mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B}\end{bmatrix}=\begin{bmatrix}0 & 0 & 1\\ 0 & 1 & -6\\ 1 & -6 & 31\end{bmatrix}$$

$$\mathbf{K}=\begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}\mathbf{B} & \mathbf{AB} & \mathbf{A}^2\mathbf{B}\end{bmatrix}^{-1}\phi(\mathbf{A})$$

Regulator Design

$$\mathbf{K} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -6 \\ 1 & -6 & 31 \end{bmatrix}^{-1} \begin{bmatrix} 199 & 55 & 8 \\ -8 & 159 & 7 \\ -7 & -43 & 117 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 6 & 1 \\ 6 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 199 & 55 & 8 \\ -8 & 159 & 7 \\ -7 & -43 & 117 \end{bmatrix} = \begin{bmatrix} 199 & 55 & 8 \end{bmatrix}$$

Find the response based on the initial conditions using MATLAB.

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Controller Design Using Pole Placement -2

Next Week …

DO YOU HAVE ANY QUESTIONS ?